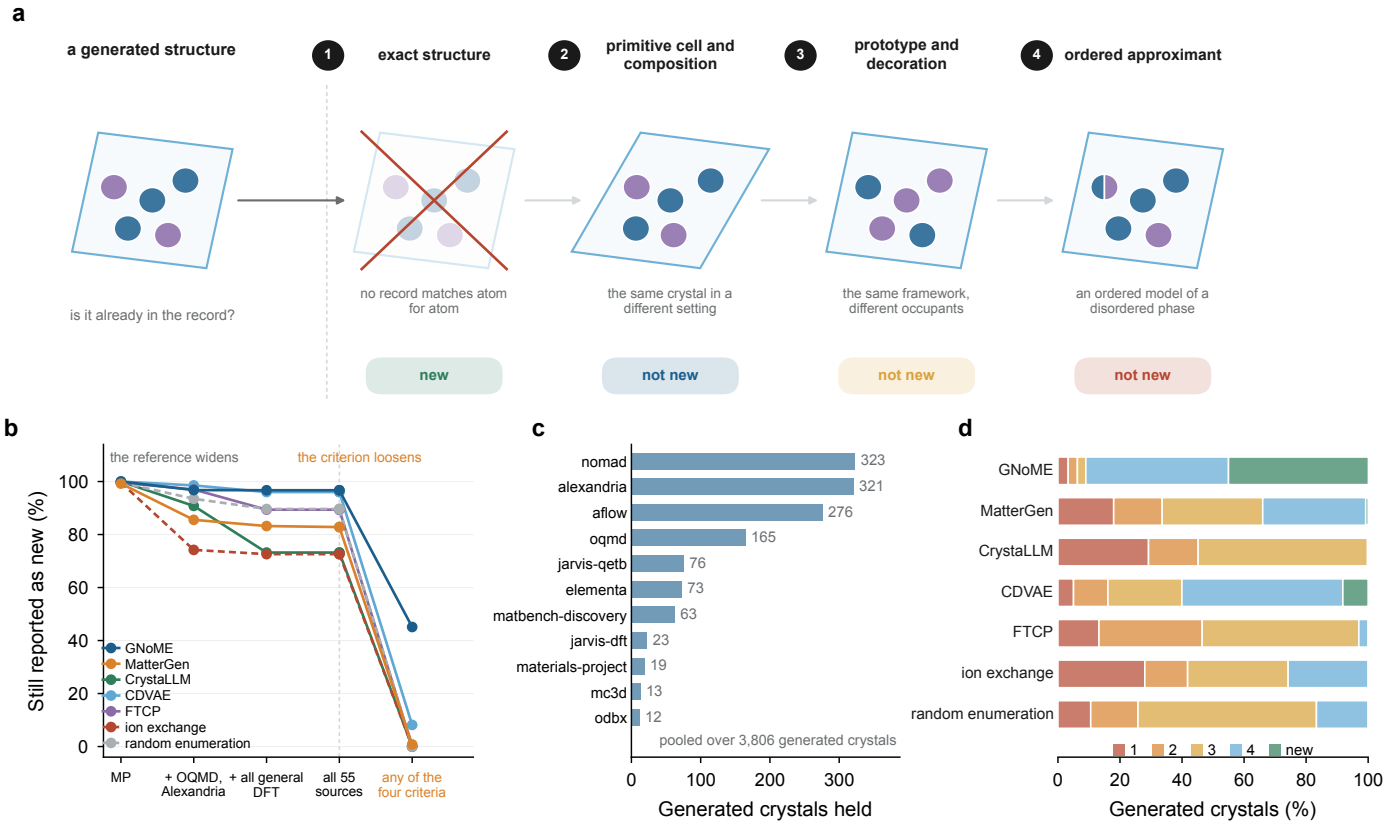
a
a generated structure
1
exact structure
2
primitive cell and composition
3
prototype and decoration
4
ordered approximant
is it already in the record?
no record matches atom for atom
the same crystal in a different setting
the same framework, different occupants
an ordered model of a disordered phase
new
not new
not new
not new
b
the reference widens
the criterion loosens
Still reported as new (%)
100
80
60
40
20
0
GNoME
MatterGen
CrystaLLM
CDVAE
FTCP
ion exchange
random enumeration
MP
+ OQMD, Alexandria
+ all general DFT
all 55 sources
any of the four criteria
c
nomad 323
alexandria 321
aflow 276
oqmd 165
jarvis-qetb 76
elementa 73
matbench-discovery 63
jarvis-dft 23
materials-project 19
mc3d 13
odbx 12
pooled over 3,806 generated crystals
0
100
200
300
Generated crystals held
d
GNoME
MatterGen
CrystaLLM
CDVAE
FTCP
ion exchange
random enumeration
1
2
3
4
new
0
20
40
60
80
100
Generated crystals (%)